WAYBIONIC

ENGINEERING SYSTEMS

Ground station software

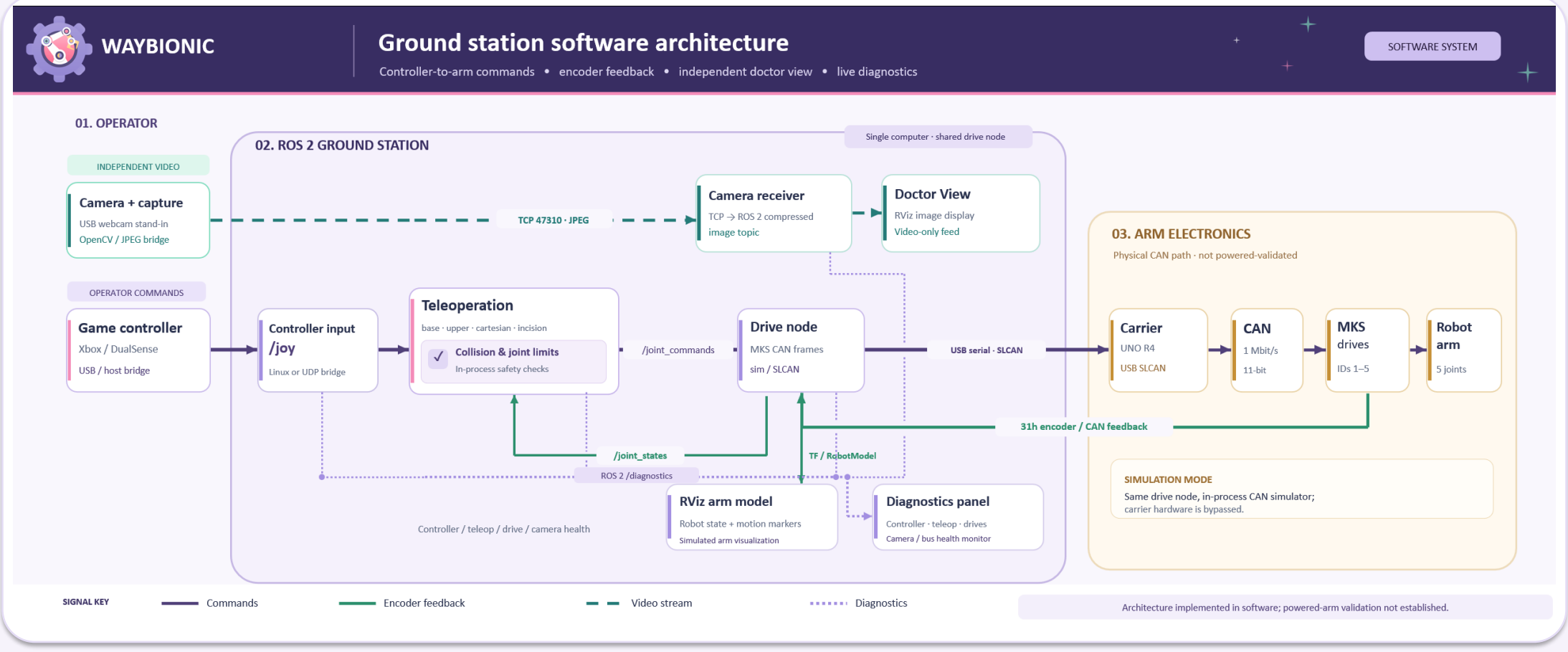
Operator control · software checks · CAN drive interface · live monitoring

SOFTWARE OVERVIEW

The ROS 2 ground station turns controller input into checked joint commands.
Video and diagnostics remain separate from arm motion.
The carrier link is software-implemented; powered five-drive validation is not documented.

01. SYSTEM ARCHITECTURE

Verified against the ROS 2 paths on main. The hardware connection is coded. It is not powered-validated.



02. TELEOPERATION + INCISION

An Xbox or DualSense controller becomes a 120 Hz stream of joint setpoints: base, upper, Cartesian, and incision. The incision group stores the tip pose when selected, then only inserts, withdraws, or tilts so the tool axis stays on that point. Sideways homing is blocked while the tip is inserted. These behaviours run in software on simulated drives, not as a clinical procedure.

03. COLLISION + JOINT LIMITS

Before a command is sent, the next pose is checked against the URDF joint ranges and collision boxes on the base, shoulder, upper arm, forearm, wrist, and tool. A step that would pass a limit, leave the tip out of reach, or deepen contact is discarded. The controller rumbles once and the label names the cause. Ranges on this model are provisional; the boxes come from the committed meshes, not a live CAD scan.

04. CAN + DRIVE FEEDBACK

The same drive node used in simulation sends MKS SERVO frames through a UNO R4 carrier: USB serial to a 1 Mbit/s CAN bus. Replies become joint states. A drive that stops answering, jumps, or lags its target stops the arm. Real drives stay still until zeroed, and stop if commands go silent. The carrier e-stop input is not wired. Software tests and virtual CAN cover this path. A powered five-drive run is not in the merged record.

05. DOCTOR VIEW + DIAGNOSTICS

A camera computer stamps each frame and sends JPEG images over TCP. RViz shows them as Doctor View and reports frame rate and capture-to-arrival delay. A webcam or a video file stands in until a procedure camera is chosen. The delay excludes time in the sensor and on the display. The monitor lists controller, teleop, drive, bus, and camera health. Video does not enter the motion-control loop.

06. SOFTWARE VALIDATION

Three results safe to print. Sources rechecked against merged PR descriptions on 27 Oct 2026.

81,421
cases, 0 mismatches
C++ vs Python frames · software only · PR #40

19 + 11
virtual-CAN tests passed
vcan host + four-node · not hardware · PR #45

30 fps
2 ms capture-to-arrival at 1280×720
Test video, not a webcam · PR #36

07. OPERATOR INTERFACE

Real captures from simulated teleop on main (cf3e887), 26 Oct 2026. No generated screenshots.



RViz simulated arm
Cartesian autoplay · simulated drives



Engineering Monitor
Live /diagnostics · simulated teleop

PENDING
Doctor View screenshot
Not captured. A team lead needs to supply a real RViz frame or an approved test-video capture.

Doctor View — pending
Do not substitute a generated image